

8-3 A

1. 求下列复合函数的偏导数(或导数):

(1) $u = f(x^2 + y^2 + z^2)$, 求 f'_x, f''_{xx}, f''_{xy} ;

(2) $z = \ln(e^x + e^y), y = x^3$, 求 $\frac{dz}{dx}$;

(3) $u = z^2 + y^2 + yz, z = \sin t, y = e^t$, 求 $\frac{du}{dt}$;

(4) $u = \arctan \frac{s}{t}, s = x + y, t = x - y$, 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$;

(5) $z = x^2y - xy^2, x = s \cdot \cos t, y = s \cdot \sin t$, 求 $\frac{\partial z}{\partial s}, \frac{\partial z}{\partial t}$;

2. 证明函数 $z = yf(x^2 - y^2)$ 满足方程

$$\frac{1}{x} \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y} = \frac{z}{y^2}.$$

3. 设 $z = f(x, y)$ 及 $x = r \cos \theta, y = r \sin \theta$ 试证下式成立:

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2.$$

4. 设函数 $f(u, v)$ 在 $\mathbb{R}^2$ 上的二阶偏导数连续, 且 $\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} = 1$, 令 $g(x, y) =$

$$f\left(xy, \frac{x^2 - y^2}{2}\right). \text{ 证明: } \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} = x^2 + y^2.$$

6. 设函数 $g(r)$ 有二阶导数, $f(x, y) = g(r), r = \sqrt{x^2 + y^2}$, 求证:

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = g''(r) + \frac{1}{r} g'(r).$$

B

1. 设 $u = f(x^2 + y^2 + z^2)$, 又 $x = r \sin \varphi \cos \theta, y = r \sin \varphi \sin \theta, z = r \cos \varphi$ 证明下列等式成立:

$$\frac{\partial u}{\partial \varphi} = 0, \quad \frac{\partial u}{\partial \theta} = 0.$$

8.4

2. 验证下列方程在指定点邻域存在以 x, y 为自变量的隐函数, 并求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

(1) $x + y - z - \cos(xyz) = 0$, 在点 $(0, 0, -1)$;

(2) $x^3 + y^3 + z^3 - 3xyz - 4 = 0$, 在点 $(1, 1, 2)$.

3. 求下列方程所确定的隐函数的导数或偏导数

(1) $\ln \sqrt{x^2 + y^2} = \arctan \frac{y}{x}$, 求 $\frac{dy}{dx}$;

(2) $z^3 - 3xyz = a^3$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$;

(3) $\cos^2 x + \cos^2 y + \cos^2 z = 1$, 求 dz ;

(4) $e^z = xyz$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}$;

(5) $x + y + z = e^{-(x+y+z)}$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$;

(6) $z^3 - 3xyz = a^2$, 求 $\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}$;

6. 设 $u(x, y) = e^{z(x, y)}$, 其中 $z(x, y)$ 是由 $ze^z = x^2 + y^2$ 确定的隐函数, 求 u'_x .

9. 设 $z = u^3 + v^3$, 而 u, v 由方程组

$$\begin{cases} x = u + v, \\ y = u^2 + v^2 \end{cases}$$

所确定, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

10. 设

$$\begin{cases} x = -u^2 + v + z, \\ y = u + vz, \end{cases}$$

求 $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}, \frac{\partial u}{\partial z}$.

13. 验证下列方程组在指定点邻域存在隐函数组, 并求所指定的全微分.

(1) $\begin{cases} u + v = x + y \\ \frac{\sin u}{\sin v} = \frac{x}{y} \end{cases}$ 在点 $(\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3})$, 求 du, dv ;

(2) $\begin{cases} x + y + z = 0, \\ x^2 + y^2 + z^2 = 1, \end{cases}$ 在点 $(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0)$, 求 $\frac{dx}{dz}, \frac{dy}{dz}$.

14. 求由下列方程组所确定的函数的导数或偏导数

$$(1) \begin{cases} z = x^2 + y^2, \\ x^2 + 2y^2 + 3z^2 = 20, \end{cases} \quad \text{求 } \frac{dy}{dx}, \frac{dz}{dx};$$

$$(3) \begin{cases} x = e^u + u \sin v, \\ y = e^u - u \cos v, \end{cases} \quad \text{求 } \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y};$$

8-7

1. 求下列曲线在指定点的切线方程与法平面方程.

$$(1) \quad x = t - \cos t, y = 3 + \sin 2t, z = 1 + \cos 3t, \text{ 点 } t = \frac{\pi}{2};$$

$$(2) \quad y = x, z = x^2, \text{ 点 } (1, 1, 1);$$

$$(3) \quad x^2 + z^2 = 10, y^2 + z^2 = 10, \text{ 点 } (1, 1, 3).$$

3. 求下列曲面在指定点的切平面方程与法线方程:

$$(1) \quad z = \arctan \frac{y}{x}, \text{ 点 } \left(1, 1, \frac{\pi}{4}\right);$$

$$(2) \quad x^2 + y^2 + z^2 = 169, \text{ 点 } (3, 4, 12);$$

$$(3) \quad x^3 + y^2 + z^3 = 1, \text{ 点 } P(-1, 1, 1);$$

8-8

1. 求下列函数的极值

$$(1) \quad z = x^3 + 3xy^2 - 15x - 12y;$$

$$(2) \quad z = x^2 + (y - 1)^2;$$

$$(3) \quad z = 6x^2 + 9y^2 + 4xy - 32x - 64y + 256;$$

$$(4) \quad z = (2ax - x^2)(2by - y^2), ab \neq 0;$$

2. 求由方程 $x^2 + y^2 + z^2 - 2x + 4y - 6z - 11 = 0$ 所确定的隐函数的极值.